Self-Distillation Inherits the Reversal Curse: What Consolidating New Knowledge from Context Does and Does Not Teach

Nov 5, 2026 · Marc Smith, Mojeek

How this was made. This study was carried out with Claude (Anthropic), which designed most of the experiments and controls, wrote the analysis code and drafted this write-up. Marc ran the experiments, decided what to test and checked the results. Predictions for the controls were written into the code before each run. How it came about: blog post. Code and result files: github.com/Mojeek/self-distillation-reversal-curse.

Abstract

A growing family of methods moves knowledge from context into weights by training a model on its own samples, scored by a teacher that holds the knowledge: sleep-style consolidation, context distillation and on-policy self-distillation. Their evaluations mostly test recall in the direction the facts were given. We ask whether reversals and two-step compositions are consolidated too, since a teacher with the facts in context answers them. In a synthetic world of invented films, directors and studios, on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, mostly they are not, for two reasons. (The student has already learnt the facts once, one way round, so this is consolidation of recently learnt knowledge, as in sleep-style systems, not learning from scratch.) First, an answer is consolidated only when the student's own text reaches the cue followed by the relation ("The Silent Winter was directed by…") and its prompt does not already make the answer available. A free passage about the entities, with their names in the prompt, states reversals correctly in about a fifth of its sentences and consolidates none of them; the same cued prompt with the names added falls from 0.85 to 0.23 on film → director, the level of free sampling; and removing film-cued prompts removes film → director but not studio → film. Second, under reverse KL, answers of a type the student was never trained to produce as a continuation stay near chance even with every cue covered; swapping which type is a continuation in the learnt facts swaps which answers fail. Forward recall stays at 0.95–1.00 throughout, so neither failure shows in the usual evaluations.

1. Introduction

This study began as a preliminary step in a broader project: giving language models something like a sleep phase, in which knowledge met during use is consolidated into their weights. Several existing methods already do a version of this, training the model on its own samples scored by a teacher that holds the knowledge: sleep-style systems [Behrouz

et al. 2026], context distillation and on-policy self-distillation [Snell et al. 2022; Padmanabhan et al. 2023; Ye et al. 2026; Shenfeld et al. 2026]. Before building on them, we wanted to know what this kind of consolidation actually transfers.

These methods are mostly evaluated on recall of facts in the direction they were given. The reversal curse says that is the easy part: a model fine-tuned on "A is the director of B" recalls B from A but not A from B, and does not combine two learnt facts [Berglund et al. 2024; Allen-Zhu & Li 2025]. With the facts in context the same model does both. A teacher with the facts in context answers reversals and compositions, so one might expect distillation from it to carry them into the weights. We test whether it does; evaluations of forward recall would not show it if it did not.

We use a teacher with the facts in context and a student without them, a close analogue of the consolidation step of Behrouz et al. (whose teacher holds the knowledge in fast parametric experts, not in context), with on-policy distillation in the style of GKD [Agarwal et al. 2024]. Our findings:

1. **The student's text must reach the cue and relation, without the answer at hand.** Free and prefix sampling add little beyond fine-tuning on the facts. Their passages state reversals correctly, but with every name listed in the prompt the student can pick the answer by recognition, already agrees with the teacher, and learns nothing that holds without the list. One prompt per entity, naming only that entity, closes most of the gap; adding the list back to it removes the whole gain.
2. **Removing a cue removes its row.** At matched volume, removing film-cued prompts returns film → director to chance and leaves studio → film intact, and removing studio-cued prompts does the reverse. The cue need not be the first word: with studio-cued prompts that may not begin with the studio's name, studio → film still reaches 0.81.
3. **Reverse KL does not consolidate answers the student is never trained to produce as continuations,** even with every cue covered. Swapping which entity type appears as a continuation in the learnt facts swaps which answers fail.
4. **The standard-world patterns replicate on Llama-3.1-8B,** three seeds each; the crossed-world, list, mid-sentence and volume-matched controls are on Qwen.

2. Setup

World. Each seed draws 40 invented films, each with one director and one studio, and each studio producing one film, so every two-step question has one answer. People, titles and studios are each matched in token length. In the standard world the facts are " {person} is the director of {film}." and "{film} was produced by {studio}.", so films and studios appear as continuations and people only start sentences. In the crossed world (Section 3.3) the facts are "{film} was directed by {person}." and "{studio} produced {film}.", so people and films appear as continuations and studios only start sentences.

Learning the facts. The student first learns the 80 fact sentences with LoRA (rank 16, alpha 32, learning rate 1e-4, one step per sentence, 12 epochs). This is the starting point for every consolidation condition, so the student already partly holds the knowledge, as in Behrouz et al., whose student starts from weights updated in the wake phase.

Teacher and student. The teacher is the base model (adapter off) with a window of five films' facts in its prompt. The student is the same model with the adapter on and no facts. On-policy distillation (OPD): the student samples a response (temperature 1.0, top-p 0.95, up to 160 tokens), and is trained on the token-level divergence from the teacher's distribution over the same tokens; forward KL unless stated. Each condition trains on 960 sequences (gradient accumulation 4, clipping 1.0), so cueing costs no more than free sampling.

Sampling conditions.

Condition	Student prompt	Answers in the prompt?
free	a short passage about the window's 15 entities, all named in random order	yes, all names listed
free, no names	a short passage about some films, directors and studios	no; the student writes about other films
prefix	continue the first 4–16 tokens of a teacher passage (free prompt)	yes
cued	one or two sentences starting with one named entity, every entity in turn (8 per entity)	no
cued, no film / no studio	as cued, without film / studio prompts (640 sequences, 8 per entity)	no
cued + list	as cued, plus the window's 15 names as in the free prompt	yes
cued, mid-sentence	as cued, but the first sentence may not begin with the entity's name	no

Off-policy comparisons (SFT) fine-tune the student on teacher-written text from the free or cued prompt, at the same 960 sequences (Qwen only).

Probes. Plain-text prompts ending where the answer should follow, with nothing in context: forward (two phrasings), film → director (two phrasings), studio → film, and two steps each way (director → studio, studio → director). The correct answer is scored against four matched distractors drawn from the other films' entities, by summed log-

probability; we report accuracy (correct ranked first; chance 0.20). References: facts only (no consolidation), facts in context (base model, the item's and distractors' facts in the prompt) and gold (facts plus four correct derived statements per film, worded unlike any probe).

Models and seeds. Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, three seeds each (seed sets the world and the sampling). Gap closed = (condition − facts only) / (in context − facts only).

3. Results

Fine-tuning on the facts alone reproduces the reversal curse and adds a composition failure: forward recall is 0.98–1.00 on both models, including a new phrasing, while film → director, studio → film and both two-step rows are at chance (0.15–0.28). With the facts in context the same models answer reversals at 0.91–0.97 and two steps at 0.61–0.79. Forward recall stays at 0.95–1.00 in every consolidation condition below, so we report only the rows that need consolidation. Intervals are 95% bootstraps over films within each seed (120 films per row); they omit between-seed variation, so we call a difference reliable only when its interval excludes zero and all three seeds agree in sign.

Table 1. Qwen2.5-7B, forward KL, accuracy (3 seeds). Free without names and prefix sampling are in Section 3.1, cued mid-sentence in Section 3.2. Last row: share of sentences in which the student states film → director correctly (the film before its own director, "direct…" between), from the stored samples.

Probe	Facts only	Free	Cued	Cued + list	Cued, no film	Cued, no studio	In context	Gold
Film → director	0.16	0.23	0.85	0.23	0.20	0.83	0.91	0.96
Film → director, new phrasing	0.17	0.24	0.87	0.22	0.17	0.78	0.93	0.94
Studio → film	0.17	0.38	0.91	0.40	0.88	0.22	0.97	0.99
Two steps, director → studio	0.26	0.33	0.67	0.38	0.72	0.53	0.72	1.00
Two steps, studio → director	0.15	0.24	0.57	0.21	0.28	0.17	0.71	0.94
Film → director stated correctly		21%	2%	52%	0%	5%		

Llama-3.1-8B (3 seeds; full table in the appendix) shows every pattern. Film → director: 0.15 facts only, 0.23 free, 0.92 cued, 0.34 with no film cues (at 960 sequences, not volume-matched), 0.22 cued under reverse KL, 0.97 in context. Studio → film: 0.18, 0.47, 0.98, 0.97,

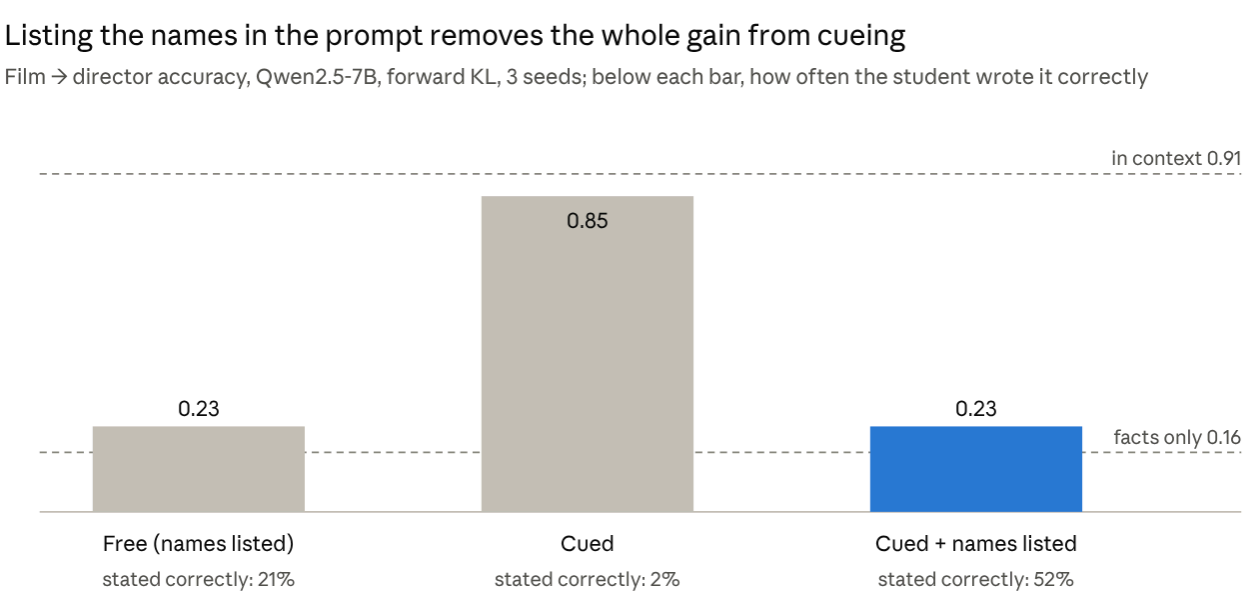
0.61 and 0.97.

3.1 Free sampling states reversals and does not learn them

Free and prefix sampling move the reversal and two-step rows by at most 0.09 over facts only, except studio → film (Qwen 0.38, Llama 0.47, from 0.17–0.18). This is not because the reversals go unwritten. The student's free passages state film → director correctly, the film before its own director with "direct…" between, in 21% of sentences on Qwen and 25% on Llama, under both divergences, and so does the teacher text used for off-policy training (23%); accuracy stays at chance in every case. The free prompt lists every entity of the window, so the student can choose a film's director by recognising one of the five listed people rather than recalling it. Its answer then already matches the teacher's, there is little to learn, and whatever is learnt is conditioned on a list the probe does not contain. Section 3.2 tests this directly. Without the names the student writes about films of its own invention and almost never about ours, so it fails for a different reason (reversals at 0.21; studio → film falls from 0.38 to 0.16, −0.23 [−0.33, −0.12]).

3.2 Reaching the cue and relation, without the answer at hand

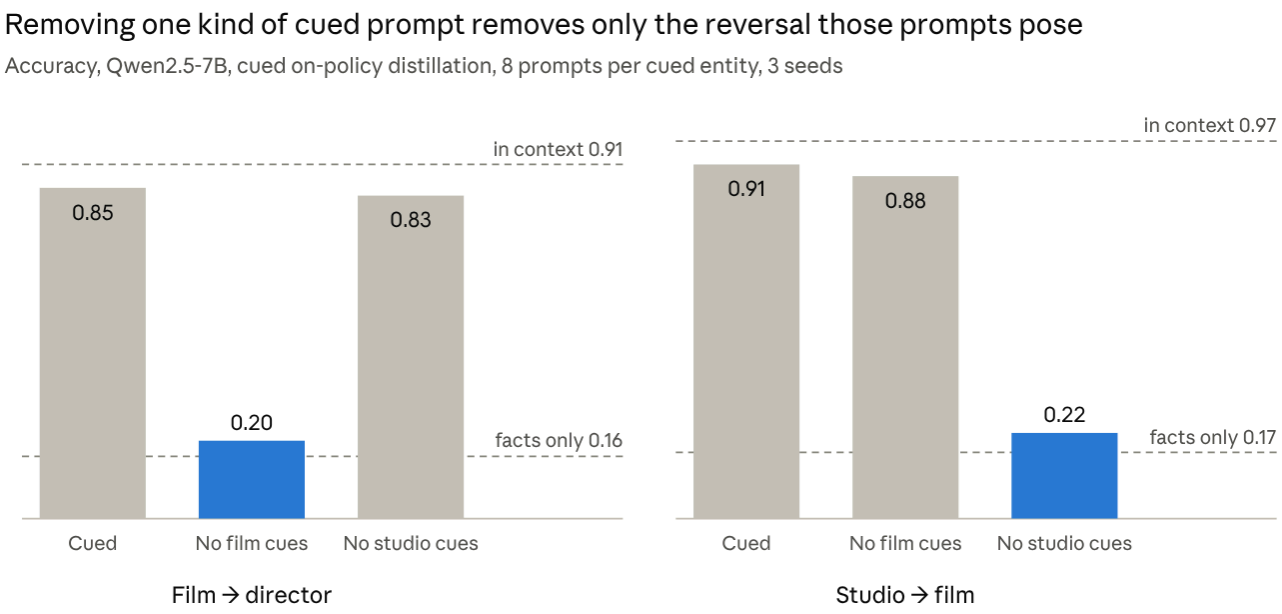
One prompt per entity, naming only that entity, closes 75–92% of the in-context gap on Qwen and 89–101% on Llama, on every reversal and two-step row (film → director: 0.85 and 0.92). Its samples state film → director correctly in only 2% of sentences, because the student mostly invents the director; the teacher, scored on the student's own tokens, corrects the guess (examples in A10). Adding the window's 15 names to the same prompt removes the whole gain: film → director 0.23, no different from free sampling (0.00 [−0.06, 0.06]), and every reversal and two-step row falls by 0.29–0.65 against plain cued, all seeds agreeing. The list barely changes where sentences start (20% begin with a film, against 26%), and its samples state film → director correctly in 52% of sentences, more than any other condition. Being right in the sample is not what consolidates the answer; having to produce it is.



Qwen2.5-7B, standard world, 3 seeds; accuracy from the result files, statement shares from order_check.py on the stored samples

Figure 2. With the names in the prompt, the student writes the reversal correctly more often and learns none of it; without them it mostly guesses and learns the reversal.

The controls show the other half: the student's text must reach the cue. At 8 prompts per entity, removing film-cued prompts returns film → director to chance (0.20 and 0.17) while studio → film stays at 0.88; removing studio-cued prompts returns studio → film to chance (0.22) while film → director stays at 0.83 and 0.78. Each control loses 0.65–0.69 on its own row (intervals 0.55–0.78) and 0.02–0.08 on the other. A later mention in another clause does not substitute: the no-studio samples name studios in clauses such as "…produced by Zantor Pictures, which also produced…", and studio → film stays at 0.22. Binding errors (sentences mixing two films, 17–26% in the controls and 78% with the list) do not explain the pattern: each control spares the other row, and cued off-policy training on text with 32% binding errors reaches 0.97.



Qwen2.5-7B, standard world, 3 seeds; accuracies from summaries.txt, matched at 8 prompts per entity

Figure 1. Removing one kind of cued prompt removes only the reversal those prompts pose (blue). Dashed lines mark facts only and facts in context.

The cue need not be the first word. With cued prompts that forbid beginning the first sentence with the entity's name, only 1% of sentences begin with a studio, yet studio → film reaches 0.81 (cued 0.91, facts only 0.17); the studio usually comes a few words in ("The renowned Zantor Pictures produced…"). The same run does not decide film → director (0.66), because the student still opened 32% of sentences with a film, often the cued person's or studio's own film.

The two-step rows fit a simple account: each hop must be consolidated in the direction it is traversed. Director → studio traverses both facts forwards and survives both controls (0.72, 0.53). Studio → director traverses both in reverse, needs studio- and film-cued prompts, and falls in both (0.28, 0.17; losses of 0.29 and 0.40, both reliable). It is also volume-sensitive: with 12 prompts per remaining entity the no-film run reached 0.68, for reasons the stored samples do not reveal (appendix). We claim the dissociation for single-hop reversals and report two-step rows descriptively.

Off-policy fine-tuning on teacher text behaves the same way: from the free prompt it stays near chance (0.18–0.35 on single-hop reversals) although the teacher's text states film → director correctly in 23% of sentences; from the cued prompt it reaches 0.85–0.97. The limit belongs to learning from sampled text whose context differs from the question's, and on-policy distillation inherits it.

The same dependence appears when the model's own notes are learnt, with no distillation: question-and-answer notes start from the film and reverse film → director (0.67) but not studio → film (0.18), in all three seeds.

3.3 Reverse KL does not consolidate answers the student is never trained to produce

All results above use forward KL. Under reverse KL, with full cueing, the rows whose answer is a person stay near chance on both models (film → director: Qwen 0.23, Llama 0.22), while studio → film keeps part of its gain (0.59, 0.61). Reverse KL penalises mistakes mainly on tokens the student already gives substantial probability [Tang & Yang 2026]. In the standard world, people never appear as continuations in the learnt facts, so the student was never trained to produce a person's name in answer position. Studios do appear as continuations there, and the studio-answer two-step row partly recovers too (director → studio: Qwen 0.53 [0.32, 0.75], Llama 0.28 [−0.00, 0.57]).

The crossed world tests this by moving people into continuation position and studios out of it. The probes are unchanged, so in this world the reversals are director → film, film → studio and director → studio, and studio → director becomes a forward composition. With forward KL every reversal is consolidated (director → film 0.93, film → studio 0.85, director → studio 0.62; in context 0.99, 0.85, 0.66). With reverse KL the failure moves to studio answers.

Table 3. Reverse KL, cued, gap closed by answer type, 95% intervals in brackets.

World	Answer type never a continuation in the facts	Film answers	Person answers, when a continuation
Standard, Qwen	person: 0.10 [−0.01, 0.21] (film → director)	0.52 [0.40, 0.63] (studio → film)	—
Standard, Llama	person: 0.08 [−0.02, 0.18] (film → director)	0.54 [0.42, 0.65] (studio → film)	—
Crossed, Qwen	studio: 0.04 [−0.09, 0.15] (film → studio), −0.04 [−0.25, 0.13] (director → studio)	0.57, 0.58 [0.45–0.69] (director → film, two phrasings)	0.49 [0.27, 0.72] (studio → director)

The predictions were written before the crossed runs. If reverse KL failed for person answers as such, studio → director would fail in the crossed world; it closes 0.49 of its gap. If it failed for reversing the director fact, director → film would fail; it closes 0.57. Neither holds. What fails is the answer type that never appears as a continuation in the learnt facts, whichever type that is. Two qualifications: film answers, a continuation in both worlds, close only about half the gap under reverse KL against nearly all under forward KL, so reverse KL costs something beyond this account; and 40% of reverse-KL student sentences in the crossed world are negations ("X did not direct…"), against 18% under forward KL. Negations do not explain the failure: in the standard world the reverse-KL

student wrote fewer negations than the forward-KL one (11% against 19%) yet failed on person answers, and in the crossed world its negations name a studio no more often than its other sentences (23% against 26%). What the stored samples do show is the predicted gap: under reverse KL the failing answer type appears almost only where the prompt makes a sentence start with it (crossed world: studios named in 26% of sentences, 26% start with one; standard world: people 34% and 30%), while under forward KL it also appears later in sentences (61% and 20%; 38% and 28%). In the crossed world the reverse-KL student writes, for example, "produced by the studio" where a name should be (appendix A10). The crossed-world contrast is large; the standard-world one is small, from 60 samples per seed. A log-probability check of the student's readiness to begin each answer type was inconclusive (its pre-stated prediction went the wrong way), and we do not use it as support.

3.4 Smaller observations

- **Neither on-policy nor off-policy is better overall.** Cued SFT on teacher text beats cued OPD on film → director (0.97 vs 0.85; +0.12, CI 0.07–0.19), and OPD beats SFT on director → studio (0.67 vs 0.54; +0.12, CI 0.03–0.22). Both hold in all three seeds; the other rows are level within noise. Cued SFT text had 32% binding errors (entities from two films in one sentence) and still reached 0.97.
- **Two steps forward reaches the in-context level, not above it.** After cued consolidation, director → studio is 0.67–0.72 on Qwen (in context 0.72) and 0.62 on Llama (0.61); every interval for the difference spans zero, and Llama's seeds disagree in sign. Gold derived statements reach 1.00, so learnt derived knowledge can exceed reasoning over the facts in context, but OPD did not show this at matched volume.
- **The gain is not probe wording.** One film → director probe's exact wording appears in 19 of 180 stored cued samples and the other's never, yet both rows score alike (0.85, 0.87); in the seed whose samples contain neither wording, cueing still closes 0.87 of the gap.

4. Related work

Reversal and composition. Models fine-tuned on "A is B" fail to answer "B is A" [Berglund et al. 2024], and inverse knowledge search fails without augmentation even when the forward facts are learnt [Allen-Zhu & Li 2025].

Augmentation. Reverse training adds each training string in reverse order, keeping entities intact [Golovneva et al. 2024]. EntiGraph synthesises text relating the entities of a small corpus before continued pretraining [Yang et al. 2025]. Deductive closure training has the model generate implications of seed documents, judge which are consistent, and fine-tune on those, with gains reported on the Reversal Curse dataset [Akyürek et al.

2024]. Lampinen et al. 2025 augment fine-tuning data with the model's in-context inferences and bring reversals near ceiling; their augmentation prompt includes examples of reversals, which our controls suggest is why it works. They also find that splitting documents into separate sentences improves fine-tuning (one unaugmented baseline excepted), which fits our account, since each sentence is then learnt without the document's other facts in context to answer it (our reading, not theirs).

Context distillation for knowledge. Snell et al. 2022 distil the effect of a prompt into weights. Padmanabhan et al. 2023 propagate injected entity knowledge by distilling, over continuations generated from the entity's definition, from the model with the definition in context into the model without it; the loss is forward KL on the tokens after the entity mention, so by construction only what follows the entity is distilled, and their probes (Entity Inferences, ECBD) are cloze sentences about the entity, with no reverse questions. Kujanpää et al. 2024 inject facts from free-form documents by self-distillation, with gains over fine-tuning and in places over retrieval. MixSD mixes tokens from the model's fact-in-context and naive conditionals to inject facts with less forgetting [Liu et al. 2026]. GATES distils document-grounded reasoning traces, gated by agreement among samples, into a student without the document [Stein et al. 2026].

On-policy distillation. GKD [Agarwal et al. 2024] and MiniLLM [Gu et al. 2024] train students on their own samples; MiniLLM uses reverse KL. On-policy context distillation moves experiential knowledge and system-prompt behaviour into weights with reverse KL, evaluated on reasoning, games and domain tasks [Ye et al. 2026]. SDFT uses the model conditioned on demonstrations as its own teacher for on-policy continual learning of skills and knowledge; its objective is written as reverse KL, but the authors report that forward KL works best in practice [Shenfeld et al. 2026]. For knowledge, the student answers generated questions about new articles while the teacher also sees the source text and a worked answer, so the questions posed are the training questions, without the answer in the student's prompt; their out-of-distribution test uses indirect questions, not reversals. Tang & Yang 2026 show, with synthetic lookup-table facts, that reverse-KL on-policy distillation transfers compositional skill but little factual knowledge, because reverse KL penalises mistakes mainly on tokens the student already gives substantial probability; forward KL restores factual transfer whether prefixes come from the student or the teacher. They do not vary query direction. We add the first condition, and identify which answers reverse KL fails on: those of a type the student was never trained to produce as a continuation, which includes reversals. Where reverse-KL methods are used to inject facts, the second failure should apply to them.

Sleep-style consolidation. Behrouz et al. 2026 consolidate a faster block's experts, which hold recent knowledge, into slower parameters by GKD-style distillation from the pre-update model, plus an RL reward for completing random prefixes of teacher-generated "dreams". The divergence is not specified, sampling continues teacher-generated dreams

with no entity cueing, and knowledge is evaluated with no-context SQuAD questions following SEAL, with no reverse or multi-step probes. Their teacher holds the knowledge parametrically, so our in-context teacher is an analogue of their consolidation step, not a reproduction.

What is new. EntiGraph, deductive closure training, Lampinen et al. and Padmanabhan et al. generate or distil derived text and train on it. None tests whether consolidation from self-generated text carries consequences the teacher can answer, or identifies what decides it: that the student's text must reach the cue and relation, that the answer must not be available in the student's prompt, and how both interact with the divergence. Tang & Yang identify the divergence effect for facts queried one way; we show which answers it fails on, and that these include reversals whose answer type never appears as a continuation. Our contribution is these conditions, with controls that identify them, not a new method.

Sources (checked 1 November 2026). Agarwal et al. 2024 · Akyürek et al. 2024 · Allen-Zhu & Li 2025 · Behrouz et al. 2026 · Berglund et al. 2024 · Golovneva et al. 2024 · Gu et al. 2024 · Kujanpää et al. 2024 · Lampinen et al. 2025 · Liu et al. 2026, MixSD · Padmanabhan et al. 2023 · Shenfeld et al. 2026 · Snell et al. 2022 · Stein et al. 2026, GATES · Tang & Yang 2026 · Yang et al. 2025, EntiGraph · Ye et al. 2026

5. Limitations and recommendations

Limitations.

- The world is small and synthetic, with two relations and a closed, known set of entities. In an open world, choosing which cues to cover is itself a problem we do not address.
- Our teacher holds the facts in context; sleep-style teachers hold them parametrically. We test an analogue of their consolidation step, not their method. Their imitation reward and GKD-style JSD were not tried; since the reward copies teacher text from random prefixes, which keeps the teacher's own questions, we expect it not to change the first failure.
- That the list works by letting the student answer by recognition is our explanation of the cued-with-list result; we did not measure the teacher–student divergence at those tokens.
- Whether the cue's position in the sentence matters is settled only for studio → film (0.81 with 1% studio starts); the mid-sentence run left film-first sentences at 32% and does not decide film → director. Our counts of cue-and-relation contexts in the samples cannot tell a question that sets up the probe's answer from a later clause that mentions the cue.

- The volume-matched controls and the crossed world are on Qwen only; Llama has the no-film control at 960 sequences.
- The account of the two-step rows is post hoc and leaves one earlier run unexplained (Section 3.2).
- The reverse-KL account rests on a manipulation of training format and on what the student writes (60 stored samples per seed), not on measuring the student's probabilities, and does not explain why film answers recover only half their gap.
- One LoRA setting; probes are plain text while consolidation runs in chat format; a few cued samples refer to "the given facts" (5 of 180 stored), leakage from the teacher's context.
- Intervals resample films within seeds and omit between-seed variation; three seeds is too few to resample.

Recommendations for anyone consolidating knowledge by self-distillation.

1. Make the student write from every cue a later query might start from through to the relation, and keep the answers out of its prompt: no lists of names, source text or hints the query will not have.
2. Use a mean-covering divergence such as forward KL; reverse KL will not consolidate answers the student does not already produce.
3. Evaluate reversals and multi-step questions; forward recall and retention cannot show either limit.

Appendix

A1. Prompts

All prompts are sent through each model's chat template. The teacher sees the same prompt preceded by a window of five films' facts, shuffled: "Here are some facts:\n{facts}\n\n{prompt}".

Condition	Student prompt
free	Write a short passage about these: {the window's 15 entities, in random order}.
free, no names	Write a short passage about some films, who directed them and which studios produced them.
prefix	The free prompt, followed by the first 4–16 tokens of a teacher passage as the start of the response

Condition	Student prompt
cued	Write one or two sentences about {e}, starting with "{e}", saying how {e} is linked to any people, films and studios you know of.
cued, no film / no studio	The cued prompt, cycling only over people and studios / people and films
cued + list	The cued prompt, then: Names you may use: {the window's 15 entities, in random order}.
cued, mid-sentence	Write one or two sentences about {e}, saying how {e} is linked to any people, films and studios you know of. Do not begin the first sentence with "{e}".

Cued prompts cycle through every eligible entity in shuffled order, each with its own film's window as the teacher's context.

A2. Facts, probes and gold statements

	Standard world	Crossed world
Director fact	{p} is the director of {t}.	{t} was directed by {p}.
Studio fact	{t} was produced by {s}.	{s} produced {t}.

Probe	Prompt	Answer
forward, trained	{p} is the director of	film
forward, new	{p} made the film	film
film → director	{t} was directed by	person
film → director, new	The director of {t} is	person
film → studio	{t} was produced by	studio
studio → film	{s} produced the film	film
director → studio	{p} directed a film produced by	studio
studio → director	The film produced by {s} was directed by	person

In the crossed world the probes are unchanged, so different rows are reversals: director → film, film → studio and director → studio. Gold statements: "{t} is a film whose director is {p}.", "{s} is the studio behind {t}.", "{p} made a film for {s}.", "{s} hired {p} to direct one of its films." Names: people are a first name plus two invented syllables, films "The {adjective} {noun}", studios "{syllable}{syllable} Pictures"; each set is matched in token length under the model's tokenizer.

A3. Hyperparameters

Setting	Value
Base model precision	bfloat16; LoRA weights fp32
LoRA	rank 16, alpha 32, dropout 0; q, k, v, o, gate, up, down projections
Learning the facts	AdamW, lr 1e-4, no weight decay; 1 step per sentence, 12 epochs over 80 sentences; gradient clipping 1.0
Consolidation	960 sequences (640 for the volume-matched controls); AdamW, lr 1e-4; gradient accumulation 4; clipping 1.0
Sampling	temperature 1.0, top-p 0.95, up to 160 new tokens, batches of 16
Divergence	forward KL, KL(teacher ‖ student), per token, averaged over the response; reverse KL, KL(student ‖ teacher)
Off-policy (SFT)	one pass over the teacher's responses, loss on the response only, same optimiser settings
World	40 films per seed, window of 5 films, 4 matched distractors per probe

A4. Llama-3.1-8B, full results

Accuracy, 3 seeds. With the facts in context, Llama scores the trained forward phrasing lower than the new one (0.73 against 0.97); no claim rests on that row. Gold derived statements reach 1.00 on every row.

Probe	Facts only	Free	Cued	Cued, no film (960)	Cued, reverse KL	In context
Forward, trained	1.00	0.99	1.00	1.00	0.99	0.73
Film → director	0.15	0.23	0.92	0.34	0.22	0.97
Film → director, new	0.17	0.24	0.88	0.33	0.21	0.97
Studio → film	0.18	0.47	0.98	0.97	0.61	0.97
Two steps, director → studio	0.28	0.32	0.62	0.80	0.38	0.61
Two steps, studio → director	0.16	0.16	0.72	0.76	0.17	0.79

A5. Reverse KL, all conditions (Qwen2.5-7B)

Standard world, accuracy:

Probe	Answer	Free	Prefix	Cued	Cued, no film (960)	Cued, forward KL
Forward, both relations	film, studio	0.98–1.00	1.00	0.99–1.00	0.98–1.00	1.00
Film → director	person	0.23	0.20	0.23	0.19	0.85
Film → director, new	person	0.23	0.21	0.25	0.20	0.87
Studio → film	film	0.40	0.34	0.59	0.68	0.91
Two steps, director → studio	studio	0.31	0.33	0.50	0.55	0.67
Two steps, studio → director	person	0.22	0.19	0.23	0.21	0.57

Crossed world, cued, accuracy:

Probe	In this world	Facts only	Forward KL	Reverse KL	In context
Director → film (two phrasings)	reversal, film answer	0.18	0.93	0.64	0.97–0.99
Film → director (two phrasings)	trained direction	0.97–1.00	0.98–0.99	0.99–1.00	0.97–0.99
Film → studio	reversal, studio answer	0.18	0.85	0.21	0.85
Studio → film	trained direction	1.00	1.00	1.00	0.95
Director → studio	double reversal, studio answer	0.18	0.62	0.17	0.66

Probe	In this world	Facts only	Forward KL	Reverse KL	In context
Studio → director	forward composition, person answer	0.29	0.65	0.49	0.70

A6. No-film control at 960 sequences (Qwen2.5-7B)

The first no-film run kept 960 sequences, so each person and studio had 12 prompts instead of 8. Film → director 0.30 (new phrasing 0.30), studio → film 0.93, two steps director → studio 0.84, studio → director 0.68. Against the volume-matched run, studio → director is 0.41 higher [0.30, 0.52] and film → director 0.10 higher. In the stored samples, about 5 of 84 studio-start sentences name that studio's director (none of 83 at 640), and 83% name its film (46% at 640).

A7. Sample checks

All counts are over the stored student samples: the last quarter of training, up to 60 per seed.

- **Probe wording.** One film → director probe's exact wording appears in 19 of 180 cued samples (13, 0 and 6 by seed); the other wording never appears.
- **Teacher leakage.** 5 of 180 cued samples refer to "the given facts".
- **Negations and answer types**, cued:

Run	Negations	Negations naming a studio / all naming a studio	Failing type named / starts with it
Crossed, reverse KL	40%	23% / 26%	studio: 26% / 26%
Crossed, forward KL	18%	53% / 61%	studio: 61% / 20%
Standard, reverse KL	11%	83% / 87%	person: 34% / 30%
Standard, forward KL	19%	71% / 83%	person: 38% / 28%

- **Stated reversals** (order_check.py): share of sentences in which the student states film → director correctly, the film before its own director with "direct…" between, negations excluded, with film → director accuracy:

Run	Stated correctly	Accuracy
Qwen free	21%	0.23
Qwen off-policy, free (teacher text)	23%	0.19
Qwen prefix	23%	0.20
Qwen cued	2%	0.85
Qwen cued + list	52%	0.23
Qwen cued, no film	0%	0.19
Llama free	25%	0.24
Llama cued	8%	0.90

Without requiring the relation word the shares rise by 0–11 points and the pattern is the same.

A8. Consolidation into a student that had not learnt the facts

Preliminary: one seed, untuned, Qwen2.5-7B, 960 sequences. With the student starting from the base model rather than from the learnt facts, free, prefix and cued on-policy distillation all left even forward recall at chance, and the student learnt to imitate the teacher's references to "the given facts". Behrouz et al. report the same weakness for plain distillation and add an imitation reward; their student also starts from weights updated in the wake phase.

A9. A failed prediction: the log-probability check

Before testing the crossed world, we predicted that after learning the facts the total probability the student gives the five candidate answers, in its cued sampling context, would rise for film and studio answers but not for person answers. It rose for all three (Qwen, change from base in nats: person +23.0 and +25.8, film +27.9 and +15.5, studio +22.9 and +18.3). Only the first token of person names fell (−3.1 and −0.7; Llama −8.3 and −2.6), but first tokens differ in kind across entity types (every film begins "The"), so we do not treat this as evidence.

A10. Example student samples (Qwen, seed 0)

Abridged, from the stored samples (late in training), checked against seed 0's facts. Per-seed estimates for every contrast are in ci.csv, released with the code.

Run	Sample	Against the facts
Free	"In the vibrant world of film production, several directors and production companies have made their mark… Brenn Dralix has directed The Paper Wire, which was produced by Vorora Pictures…"	Correct, but every name is in the student's prompt, so it can be written by recognition
Cued	"The Silent Winter was directed by Torval Veylon and produced by Torlu Pictures."	Studio correct; director invented (true: Rowan Fepnim). The row still reaches 0.85, because the student is trained on the teacher's distribution over its own tokens
Cued	"Dessa Lixvey is the director of The Scarlet Letter…"	Binding error: Dessa Lixvey directed The Scarlet Island; The Scarlet Letter is Dessa Veypex's
Off-policy, cued (teacher text)	"The Burning Compass was produced by Zantor Pictures and directed by Ivor Lixlu, who is also the director of The Broken Hour."	First clause correct; second wrong (The Broken Hour is Quilla Lixvor's)
Cued, no film	"Tellix Pictures produced The Hollow Highway, which was directed by Rowan Lefex…"	Wrong film (Tellix produced The Hollow Island); director and further films invented
Cued, no studio	"The Quiet Compass was directed by Elio Dranor and produced by Mosvey Pictures… not provided in the given information."	Studio correct; director invented (true: Corvin Rindun); a reference to the teacher's context
Reverse KL, standard	"The Silent Winter was produced by Torlu Pictures and directed by… Dessa Lixnir…"	Studio correct; the person answer is invented
Reverse KL, crossed	"The Silent Winter was directed by Rowan Fepnim and produced by the studio…"	Director correct (people are continuations in this world); the studio is not named. All three stored seed-0 samples in this run write "produced by the studio"